



A comparative study of two inverse dynamic models of 6 degree-of-freedom rotary Stewart-Gough parallel robot

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Abstract

6-RSS Stewart-Gough parallel robot consists of six articulated legs that connect its base and movable plates, forming a fully parallel 6DOF manipulator. The general structure of this manipulator is studied in this paper. The main contribution of this paper is deriving two complete inverse dynamic models (IDMs) in full details by employing both Lagrangian and Newton–Euler approaches for obtaining the torques applied by the active joint actuators to achieve the desired trajectory and orientation of the movable plate. Both IDMs have been derived with the assumption that each rod's mass is homogeneous and condensed in its extreme points. The models have been validated by comparing their results with the results of the model built using Simscape multi-body for three scenarios. Results revealed that the relative root-mean-square error is better than 0.57% and 0.65%, respectively, for the Lagrangian model and the Newton–Euler model. While the error of both models is within the same range, the advantage of the Newton–Euler approach is that its calculations are faster than that of the Lagrangian approach in the simulation environment, making it probably more applicable in real-time applications, and that it results in a closed-form model making it more appropriate to be linearized and employed for linear control systems.

Keywords Parallel manipulators · Rotary Stewart–Gough robot · Complete inverse dynamic model · Lagrangian approach · Newton–Euler approach · D-H method

1 Introduction

Parallel robots are advantageous over serial robots in term of more stiffness, higher load-to-mass ratio and higher speed and precision [1]. Stewart–Gough robot, introduced by Gough [2] as a tyre tester and by Stewart [3] as a flight simulator, is a commonly used parallel robot. This robot consists of a base and movable plate, connected in-between by six articulated legs in a way that makes the robot a fully parallel-actuated 6DOF robot.

The applications of Stewart-Gough robot include active vibration rejection [4–7], position control and stabilization [8, 9], CNC machining [10–12], 6DOF sensing of forces and torques [13–15], 6DOF displacement measurement [16, 17], laparoscopy [18–21] and robot-aided diagnosis [22].

Two configurations of this robot exist. The most popular is the prismatic configuration, in which the lengths of the limbs are directly controlled by 6 hydraulic actuators or 6 electrical motors with ball-screws. The other is the rotary configuration that was initially proposed by Hunt [23] as an instance of 6DOF parallel manipulators having revolute motors. In this configuration, the limbs' lengths are indirectly controlled through 6 crank-rod articulated connections.

Since late 1980s, the rotary configuration has gained more focus from the scientific community. This is because of its advantages over the prismatic configuration. This includes achieving higher speed and dynamic properties and higher load-to-mass ratio of its movable parts through installing the rotary actuators stationary on the base plate [24]. This motivated us to choose to investigate this configuration in this work.

Having complete and precise kinematic and dynamic models of a robot is crucially important in obtaining appropriate control performance from it [25]. A 6-URS parallel robot has been considered in [26] to derive its inverse kinematic and dynamic models, in which the inverse dynamic

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model (IDM) has been obtained using the natural orthogonal component approach. In [27], the kinematic model of a 6-RSS parallel robot has been derived, its error has been analyzed and a calibration algorithm has been presented to achieve the optimal error parameters of the kinematic model. In [28], the inverse kinematic model (IKM) of a 6-RUS parallel robot has been obtained using the natural coordinates, and the general form of the IDM has been presented without providing any details about deriving it. The IKM of the same 6-RUS structure has been derived in [29, 30], where the model has been employed in a neuro-fuzzy adaptive controller in [30]. The IDM of a 6-RSS Stewart mechanism has been presented in [31] by applying the virtual work principle. This model has been employed in a close-loop identification algorithm based on a visual sensor for estimating the dynamic parameters of the robot. The IKM and trajectory planning have been derived in [32]. The IKM and differential kinematic control of a 6 DOF Hunt-type parallel robot have been obtained in [33]. The IKM and IDM of a 6-RSS parallel robot have been obtained in [34] using single link approximation in addition to homogenous mass distribution of the movable plate and rods on the cranks for simplifying the derivation of a decoupled IDM. In this approximation, every crank was assumed to be holding exactly one sixth of the weight of the moving plate along with the weight of the rod connected to it. Obviously, this assumption can only be valid when the moving plate is parallel to the base plate with no translations along the axes of the plane of the base plate, making it a coarse and erroneous approximation when the moving plate moves otherwise. The same configuration has been investigated in [35]. The same decoupled approximation has also been employed while deriving the robot's IDM in this reference. A decoupled IDM of a 6-RSS parallel robot has been derived in [36] using the force distribution algorithm (FDA) to obtain the distribution of the forces applied by the moving plate and the rods on the cranks at every moment while taking the position and orientation of the movable plate into consideration.

In this paper, we consider deriving the IDMs of the general 6-RSS Stewart-Gough robot in details using the two classical approaches, i.e. the Lagrange and Newton–Euler models. The derived models are validated through comparison with the simulation results obtained from the model developed using Simscape multi-body that formulates and solves the dynamics of the complete mechanical structure in MATLAB Simulink environment.

The remainder of this paper is structured as follows. The system description is presented in Sect. 2. The IKM and the Jacobian are derived in Sect. 3, and the complete IDMs using Lagrange and Newton–Euler formulations are derived in Sect. 4. Model validation is done in Sect. 5. The paper ends with the conclusions in Sect. 6.

2 System description

This robot, whose schematics is shown in Fig. 1, consists of a movable plate and a base connected in-between by six identical kinematic chains. The movable plate's motion in 6 degrees of freedom is initiated by six identical servomotors actuating 6 identical arms, each of which consists of a crank-rod linkage with a passive spherical joint in between. Each rod connects to the movable plate by a passive spherical joint. Therefore, each limb is an RSS (Revolute-Spherical-Spherical) kinematic chain, in which the revolute joint with the base plate is the active joint through a servomotor.

3 Kinematics

Kinematics is studying the motion while not considering the forces and torques causing it. Figure 2 depicts one kinematic chain in the investigated rotary Stewart–Gough robot. The angle ϕ_{b_i} and the distance e_i determine the position and orientation of the $\{i_0\}$ coordinate system w.r.t. the base coordinate system $\{B\}$. The angle λ_{p_i} and the distance b_i determine the position and orientation of the $\{i_7\}$ coordinate system w.r.t. the movable plate coordinate system $\{P\}$.

3.1 The inverse kinematic model

In the IKM, the joint variables are determined as functions of the position and orientation of the movable plate, which are considered to be known and given. The Denavit-Hartenberg (D-H) notation is used in this work to describe the link parameters. The Denavit-Hartenberg table corresponding to this kinematic chain is listed in Table 1.

In this table, l_1 is the crank's length and l_2 is the rod's length.

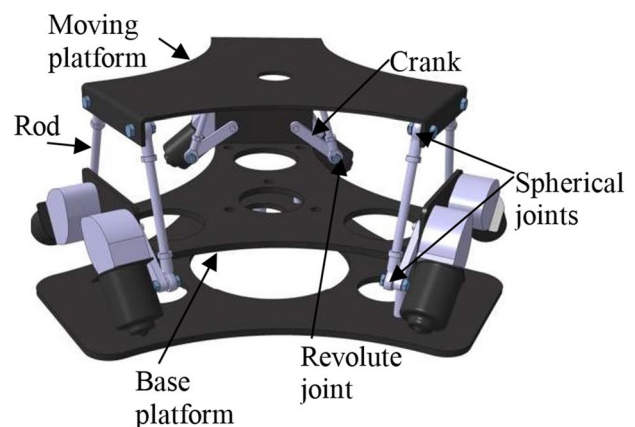


Fig. 1 Schematics of rotary Stewart-Gough robot

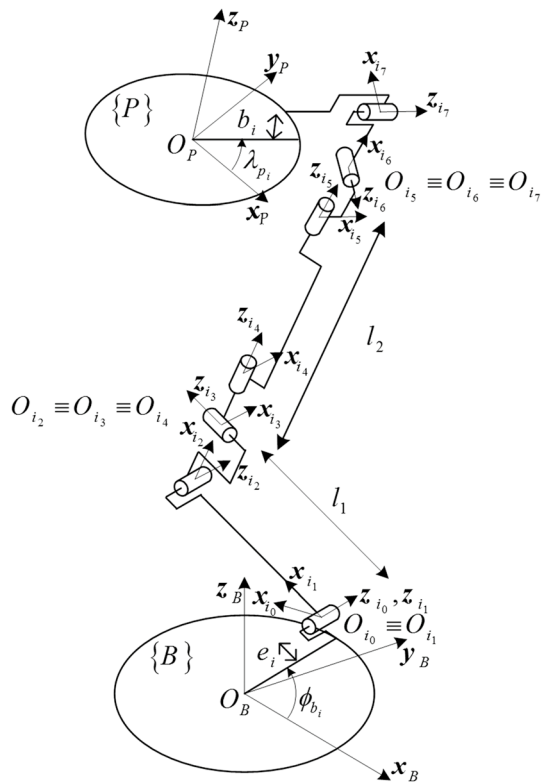


Fig. 2 The i -th kinematic chain of rotary Stewart-Gough robot

The following closure constraint relation corresponding to the i -th kinematic chain is used in deriving the IKM.

$$\overline{O_{i_0}O_{i_2}} + \overline{O_{i_2}O_{i_7}} = \overline{O_{i_0}O_B} + \overline{O_BO_P} + \overline{O_PO_{i_7}} \quad (1)$$

According to Fig. 2, the homogeneous transformation matrices ${}^B\mathbf{T}_{i_0}$ and ${}^P\mathbf{T}_{i_7}$ are as follows:

$${}^P\mathbf{T}_{i_7} = \begin{bmatrix} -s\lambda_{p_i} & 0 & c\lambda_{p_i} & r_p c\lambda_{p_i} + b_i s\lambda_{p_i} \equiv x_{rpi} \\ c\lambda_{p_i} & 0 & s\lambda_{p_i} & r_p s\lambda_{p_i} - b_i c\lambda_{p_i} \equiv y_{rpi} \\ 0 & 1 & 0 & z_{rp} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2)$$

$${}^B\mathbf{T}_{i_0} = \begin{bmatrix} -s\phi_{b_i} & 0 & c\phi_{b_i} & r_b c\phi_{b_i} + e_i s\phi_{b_i} \\ c\phi_{b_i} & 0 & s\phi_{b_i} & r_b s\phi_{b_i} - e_i c\phi_{b_i} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

in which, s and c are short hands for the sine and cosine functions respectively, and ϕ_{b_i} , e_i , λ_{p_i} and b_i are defined in Fig. 2.

The homogeneous transformation matrix ${}^B\mathbf{T}_P$ is a function of Euler angles and the displacement of the movable plate frame's center w.r.t. the base plate. This matrix is the following for yawing, pitching then rolling:

Table 1 The D-H parameters of the i -th kinematic chain of 6-RSS Stewart-Gough robot

j	a_j	α_j	d_j	θ_j
1	0	0	0	θ_i^*
2	l_1	0	0	α_i
3	0	$\pi/2$	0	$\beta_i + \pi/2$
4	0	$\pi/2$	0	γ_i
5	0	0	l_2	η_i
6	0	$\pi/2$	0	$\xi_i + \pi/2$
7	0	$\pi/2$	0	ζ_i

* Active joint variable

$${}^B\mathbf{T}_P = \begin{bmatrix} c\varphi c\psi & s\varphi s\theta c\psi - c\varphi s\psi & c\varphi s\theta c\psi + s\varphi s\psi & X_p \\ c\theta s\psi & s\varphi s\theta s\psi + c\varphi c\psi & c\varphi s\theta s\psi - s\varphi c\psi & Y_p \\ s\theta & s\varphi c\theta & c\varphi c\theta & Z_p \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

where φ , θ and ψ are roll, pitch and yaw Euler angles, respectively, and X_p , Y_p and Z_p are the components of $\overline{O_BO_P}$ in the base plate coordinate system.

Hence, the right side of Eq. (1) is the first three elements of the fourth column of ${}^B\mathbf{T}_P \cdot {}^P\mathbf{T}_{i_7} \equiv \mathbf{T}$, i.e.

$$\begin{aligned} \mathbf{T}(1,4) &= e_i + (r_p s\lambda_{p_i} - b_i c\lambda_{p_i}) [c\phi_{b_i}(c\varphi c\psi + s\varphi s\theta s\psi) + s\phi_{b_i}(c\varphi s\psi - s\varphi s\theta c\psi)] \\ &\quad - z_{rp} [c\phi_{b_i}(s\varphi c\psi - c\varphi s\theta s\psi) + s\phi_{b_i}(s\varphi s\psi + c\varphi s\theta c\psi)] \\ &\quad + Y_p c\phi_{b_i} - X_p s\phi_{b_i} - (r_p c\lambda_{p_i} + b_i s\lambda_{p_i}) s(\phi_{b_i} - \psi) c\theta \\ \mathbf{T}(2,4) &= Z_p - (r_p c\lambda_{p_i} + b_i s\lambda_{p_i}) s\theta + z_{rp} c\varphi c\theta + (r_p s\lambda_{p_i} - b_i c\lambda_{p_i}) s\varphi c\theta \\ \mathbf{T}(3,4) &= z_{rp} [c\phi_{b_i}(s\varphi s\psi + c\varphi c\psi s\theta) - s\phi_{b_i}(c\psi s\varphi - c\varphi s\psi s\theta)] \end{aligned} \quad (4)$$

The homogeneous transformation matrices involved in the left side of Eq. (1) are as follows based on the D-H table.

$$\begin{aligned} {}^{i_0}\mathbf{T}_{i_1} &= \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^{i_1}\mathbf{T}_{i_2} = \begin{bmatrix} c\alpha_i & -s\alpha_i & 0 & l_1 \\ s\alpha_i & c\alpha_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ {}^{i_2}\mathbf{T}_{i_3} &= \begin{bmatrix} -s\beta_i & -c\beta_i & 0 & 0 \\ 0 & 0 & -1 & 0 \\ c\beta_i & -s\beta_i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^{i_3}\mathbf{T}_{i_4} = \begin{bmatrix} c\gamma_i & -s\gamma_i & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\gamma_i & c\gamma_i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ {}^{i_4}\mathbf{T}_{i_5} &= \begin{bmatrix} c\eta_i & -s\eta_i & 0 & 0 \\ s\eta_i & c\eta_i & 0 & 0 \\ 0 & 0 & 1 & l_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned} \quad (5)$$

Since $O_{i_5} \equiv O_{i_6} \equiv O_{i_7}$ as depicted in Fig. 2, the left side of Eq. (1) equals the first three elements of the fourth column of ${}^{i_0}\mathbf{T}_{i_1} \cdot {}^{i_1}\mathbf{T}_{i_2} \cdot {}^{i_2}\mathbf{T}_{i_3} \cdot {}^{i_3}\mathbf{T}_{i_4} \cdot {}^{i_4}\mathbf{T}_{i_5}$. Hence, the IKM of Eq. (1) is rewritten as follows:

$$l_1 c\theta_i + l_2 c(\theta_i + \alpha_i) c\beta_i = \mathbf{T}(1, 4) \quad (6)$$

$$l_1 s\theta_i + l_2 s(\theta_i + \alpha_i) c\beta_i = \mathbf{T}(2, 4) \quad (7)$$

$$l_2 s\beta_i = \mathbf{T}(3, 4) \quad (8)$$

β_i is directly calculated from Eq. (8):

$$\beta_i = \arcsin \frac{\mathbf{T}(3, 4)}{l_2}, \quad -\frac{\pi}{2} \leq \beta_i \leq \frac{\pi}{2} \quad (9)$$

$c\alpha_i$ is obtained by squaring and summing Eqs. (6), (7) and (8). The sign of α_i is determined by noting that it can only be changed if the chain goes through a singularity case, in which $\alpha_i = 0$. This singular case is visualized by the alignment of the crank and rod. Thus, the sign of α_i is always equal to its initial sign. According to the initial configuration considered, α_i is obtained to be:

$$\alpha_i = (-1)^i \arccos \frac{\mathbf{T}(1, 4)^2 + \mathbf{T}(2, 4)^2 + \mathbf{T}(3, 4)^2 - l_1^2 - l_2^2}{2l_1 l_2 c\beta_i} \quad (10)$$

Expanding Eq. (7), substituting $c\theta_i = [1 - \tan^2(\theta_i/2)]/[1 + \tan^2(\theta_i/2)]$ and $s\theta_i = 2 \tan(\theta_i/2)/[1 + \tan^2(\theta_i/2)]$, and solving for θ_i give:

$$\theta_{i1,2} = 2 \arctan \frac{A_i \pm \sqrt{A_i^2 + B_i^2 - \mathbf{T}(2, 4)^2}}{\mathbf{T}(2, 4) + B_i}, \quad -\pi \leq \theta_i \leq \pi \quad (11)$$

where $A_i = l_1 + l_2 c\beta_i$ and $B_i = l_2 c\beta_i s\alpha_i$. By evaluating the two solutions θ_{i1} and θ_{i2} against Eq. (6), only one solution satisfies it. Thus, the inverse kinematic problem has a unique solution for all the active and considered passive joints.

3.2 Velocity analysis (Jacobian)

In the Jacobian analysis of this parallel robot, the input is $\dot{\mathbf{x}} = [\dot{\mathbf{v}}_P^T \dot{\mathbf{w}}_P^T]^T$ and the output is $\dot{\boldsymbol{\theta}} = [\dot{\theta}_1 \dots \dot{\theta}_6]^T$, where $\dot{\mathbf{v}}_P$ is the linear velocity and $\dot{\mathbf{w}}_P$ is the angular velocity of the movable plate. All other angular velocities are passive. By rewriting the closure constraint equation for the i -th limb in the form $\overline{O_B O_P} + \overline{O_P O_{i7}} = \overline{O_B O_{i0}} + \overline{O_{i0} O_{i2}} + \overline{O_{i2} O_{i7}}$ and getting the time derivative in the base frame $\{B\}$, we get:

$$\dot{\mathbf{v}}_P + \dot{\mathbf{w}}_P \times \overline{O_P O_{i7}} = 0 + \dot{\mathbf{w}}_{i1} \times \overline{O_{i0} O_{i2}} + \dot{\mathbf{w}}_{i4/i_1} \times \overline{O_{i2} O_{i7}} \quad (12)$$

We dot-multiply both sides of Eq. (12) by $\overline{O_{i2} O_{i7}}$ to remove the passive rates:

$$\overline{O_{i2} O_{i7}} \cdot \dot{\mathbf{v}}_P + \dot{\mathbf{w}}_P \cdot (\overline{O_P O_{i7}} \times \overline{O_{i2} O_{i7}}) = \dot{\mathbf{w}}_{i1} \cdot (\overline{O_{i0} O_{i2}} \times \overline{O_{i2} O_{i7}}) \quad (13)$$

where $\dot{\mathbf{w}}_{i1}$ is the angular velocity of the i -th limb's crank. By performing some algebraic calculations and combining the six equations of the six limbs in one equation, we have the velocity matrix equation, through which we can obtain the active joint rates.

$$\mathbf{J}_x \cdot \dot{\mathbf{x}} = \mathbf{J}_\theta \cdot \dot{\boldsymbol{\theta}}, \quad \text{or } \dot{\boldsymbol{\theta}} = \mathbf{J} \cdot \dot{\mathbf{x}} \quad (14)$$

where $\mathbf{J} = \mathbf{J}_\theta^{-1} \cdot \mathbf{J}_x$, $\mathbf{J}_\theta = \text{diag}(l_1 s\alpha_1 c\beta_1, \dots, l_1 s\alpha_6 c\beta_6)$ and

$$\mathbf{J}_x = \begin{bmatrix} J_{11} & \dots & J_{16} \\ \vdots & \ddots & \vdots \\ J_{61} & \dots & J_{66} \end{bmatrix}, \quad \text{in which,}$$

$$J_{i1} = c\phi_{b_i} s\beta_i - s\phi_{b_i} c(\theta_i + \alpha_i) c\beta_i$$

$$J_{i2} = s\phi_{b_i} s\beta_i + c\phi_{b_i} c(\theta_i + \alpha_i) c\beta_i$$

$$J_{i3} = s(\theta_i + \alpha_i) c\beta_i$$

$$J_{i4} = {}^B y_{rp_i} s(\theta_i + \alpha_i) c\beta_i - {}^B z_{rp_i} [s\phi_{b_i} s\beta_i + c\phi_{b_i} c(\theta_i + \alpha_i) c\beta_i]$$

$$J_{i5} = {}^B z_{rp_i} [c\phi_{b_i} s\beta_i - s\phi_{b_i} c(\theta_i + \alpha_i) c\beta_i] - {}^B x_{rp_i} s(\theta_i + \alpha_i) c\beta_i$$

$$J_{i6} = {}^B x_{rp_i} [s\phi_{b_i} s\beta_i + c\phi_{b_i} c(\theta_i + \alpha_i) c\beta_i] - {}^B y_{rp_i} [c\phi_{b_i} s\beta_i - s\phi_{b_i} c(\theta_i + \alpha_i) c\beta_i] \quad (15)$$

where,

$$\begin{bmatrix} {}^B x_{rp_i} \\ {}^B y_{rp_i} \\ {}^B z_{rp_i} \end{bmatrix} = {}^B \overline{O_P O_{i7}} = {}^B \mathbf{R} \cdot {}^P \overline{O_P O_{i7}} \\ = \begin{bmatrix} x_{rp_i} c\theta c\psi + y_{rp_i} (s\phi s\theta c\psi - c\phi s\psi) + z_{rp_i} (s\phi s\psi + c\phi s\theta c\psi) \\ x_{rp_i} c\theta s\psi + y_{rp_i} (s\phi s\theta s\psi + c\phi c\psi) + z_{rp_i} (c\phi s\theta s\psi - s\phi c\psi) \\ y_{rp_i} s\phi c\theta - x_{rp_i} s\theta + z_{rp_i} c\phi c\theta \end{bmatrix} \quad (16)$$

We consider Eq. (12) for calculating the passive joint rates, α_i and β_i . We rearrange it and express it in the $\{i_4\}$ frame to have the following:

$$\begin{aligned} {}^{i_4} \overline{\mathbf{w}}_{i4/i_1} \times {}^{i_4} \overline{O_{i2} O_{i7}} \\ = {}^B \mathbf{R} ({}^B \dot{\mathbf{v}}_P + {}^B \dot{\mathbf{w}}_P \times \overline{O_P O_{i7}}) - {}^{i_4} \mathbf{R} ({}^{i_1} \overline{\mathbf{w}}_{i1} \times {}^{i_1} \overline{O_{i0} O_{i2}}) \end{aligned} \quad (17)$$

where,

$${}^{i_4} \overline{\mathbf{w}}_{i4/i_1} = \begin{bmatrix} 0 \\ 0 \\ \dot{\gamma}_i \end{bmatrix} + {}^{i_4} \mathbf{R} \begin{bmatrix} 0 \\ 0 \\ \dot{\beta}_i \end{bmatrix} + {}^{i_4} \mathbf{R} \begin{bmatrix} 0 \\ 0 \\ \dot{\alpha}_i \end{bmatrix} = \begin{bmatrix} \dot{\alpha}_i c\beta_i \\ \dot{\beta}_i \\ \dot{\gamma}_i + \dot{\alpha}_i s\beta_i \end{bmatrix} \quad (18)$$

Thus, the left side of Eq. (17) equals to:

$${}^{i_4} \overline{\mathbf{w}}_{i4/i_1} \times {}^{i_4} \overline{O_{i2} O_{i7}} = \begin{bmatrix} \dot{\alpha}_i c\beta_i \\ \dot{\beta}_i \\ \dot{\gamma}_i + \dot{\alpha}_i s\beta_i \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ l_2 \end{bmatrix} = \begin{bmatrix} l_2 \dot{\beta}_i \\ -l_2 \dot{\alpha}_i c\beta_i \\ 0 \end{bmatrix} \quad (19)$$

Thus, $\dot{\alpha}_i$ and $\dot{\beta}_i$ are obtained by solving Eq. (17).

3.3 Acceleration analysis

We take the time derivative of Eq. (14) to calculate the angular accelerations $\ddot{\theta}_i$ of the active joints.

$$\dot{\mathbf{J}}_{\mathbf{x}} \cdot \dot{\mathbf{x}} + \mathbf{J}_{\mathbf{x}} \cdot \ddot{\mathbf{x}} = \dot{\mathbf{J}}_{\mathbf{\theta}} \cdot \dot{\mathbf{\theta}} + \mathbf{J}_{\mathbf{\theta}} \cdot \ddot{\mathbf{\theta}} \quad (20)$$

Hence,

$$\ddot{\mathbf{\theta}} = \mathbf{J}_{\mathbf{\theta}}^{-1} \cdot (\dot{\mathbf{J}}_{\mathbf{x}} \cdot \dot{\mathbf{x}} + \mathbf{J}_{\mathbf{x}} \cdot \ddot{\mathbf{x}} - \dot{\mathbf{J}}_{\mathbf{\theta}} \cdot \dot{\mathbf{\theta}}) \quad (21)$$

4 The inverse dynamic modeling

In this section, two IDMs are derived. The first one is the Lagrange model and the second one is the Newton–Euler model.

4.1 The Lagrangian inverse dynamic model

Due to the holonomic constraints involved in this parallel robots, the first type Lagrangian equations are

$$\Gamma_i = [X_p - x_{rbi} + l_1 s\phi_{b_i} c\theta_i + x_{rp_i} c\theta c\psi + y_{rp_i} (s\phi s\theta c\psi - c\phi s\psi) + z_{rp_i} (s\phi s\psi + c\phi s\theta c\psi)]^2 \\ + [Y_p - y_{rbi} - l_1 c\phi_{b_i} c\theta_i + x_{rp_i} c\theta s\psi + y_{rp_i} (s\phi s\theta s\psi + c\phi c\psi) + z_{rp_i} (c\phi s\theta s\psi - s\phi c\psi)]^2 \\ + [Z_p - l_1 s\theta_i - x_{rp_i} s\theta + y_{rp_i} s\phi c\theta + z_{rp_i} c\phi c\theta]^2 - l_2^2 \quad (25)$$

considered. This model treats the constraints as distinct equations. This is done by employing six independent generalized coordinates $\theta_i, i = 1 \dots 6$ and six redundant coordinates $X_p, Y_p, Z_p, \phi, \theta$, and ψ . Therefore, the generalized coordinates are considered to be $q = [X_p \ Y_p \ Z_p \ \phi \ \theta \ \psi \ \theta_1 \ \theta_2 \ \theta_3 \ \theta_4 \ \theta_5 \ \theta_6]^T$.

To simplify the derivation, two assumptions are considered. First, each rod's mass, m_2 , is homogenous and condensed in its extreme points, O_{i_2} and O_{i_7} . Second, the mass condensed at point O_{i_7} is dealt with as a part of the movable plate's mass, because this point is fix with the movable plate.

The first type Lagrangian equations are expressed as follows:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = Q_j + \sum_{i=1}^k \lambda_i \frac{\partial \Gamma_i}{\partial q_j}, \quad j = 1 \dots 12 \quad (22)$$

in which, Γ_i is the i -th constraint function, k is the constraint functions' number ($k = 6$ in this robot), λ_i is the i -th Lagrangian coefficient, Q_j is the j -th non-conservative torque exerted on the robot, and L is the Lagrangian function.

The non-conservative torques are the torques of the rotary actuators Q_{m_j} and of the viscous friction at active joints Q_{vf_j} .

The Lagrangian equations shown in Eq. (22) are arranged in two groups. In the first group ($j = 1 \dots 6$), the Lagrangian coefficients are the sole unknowns. These k equations relate to the redundant coordinates. In the second group, the torques of the actuators are the unknowns. This equation set relates to the actuated joint variables.

The constraint functions express the closure conditions for the robot's 6 articulated legs as follows:

$$\Gamma_i = \|O_{i_2} O_{i_5}\|^2 - l_2^2 = 0 \quad (23)$$

We have

$${}^B \overrightarrow{O_{i_2} O_{i_5}} = \begin{bmatrix} l_1 s\phi_{b_i} c\theta_i \\ -l_1 c\phi_{b_i} c\theta_i \\ -l_1 s\theta_i \end{bmatrix} - \begin{bmatrix} x_{rbi} \\ y_{rbi} \\ 0 \end{bmatrix} + \begin{bmatrix} X_p \\ Y_p \\ Z_p \end{bmatrix} \\ + \begin{bmatrix} x_{rp_i} c\theta c\psi + y_{rp_i} (s\phi s\theta c\psi - c\phi s\psi) + z_{rp_i} (s\phi s\psi + c\phi s\theta c\psi) \\ x_{rp_i} c\theta s\psi + y_{rp_i} (s\phi s\theta s\psi + c\phi c\psi) + z_{rp_i} (c\phi s\theta s\psi - s\phi c\psi) \\ y_{rp_i} s\phi c\theta - x_{rp_i} s\theta + z_{rp_i} c\phi c\theta \end{bmatrix} \quad (24)$$

Hence,

By definition, the Lagrangian function equals $L = K - U$.

K is the robot's total kinetic energy, defined as

$$K = K_p + \sum_{i=1}^6 (K_{i_1} + K_{i_2}).$$

U is the robot's total potential energy, defined as

$$U = U_p + \sum_{i=1}^6 (U_{i_1} + U_{i_2}).$$

The movable plate's kinetic energy equals

$$K_p = \frac{m'_p}{2} (\dot{X}_p^2 + \dot{Y}_p^2 + \dot{Z}_p^2) + \frac{I'_{px}}{2} (\dot{\phi} - \dot{\psi} s\theta)^2 \\ + \frac{I'_{py}}{2} (\dot{\theta} c\phi + \dot{\psi} s\phi c\theta)^2 + \frac{I'_{pz}}{2} (\dot{\theta} s\phi - \dot{\psi} c\phi c\theta)^2 \quad (26)$$

where m'_p , I'_{px} , I'_{py} and I'_{pz} are the mass and moments of inertia of the assembly of the movable plate and the parts of the rods condensed at points O_{i_7} . Hence $m'_p = m_p + 3m_2$.

The movable plate's potential energy equals:

$$U_p = m_p g Z_p \quad (27)$$

The i -th crank's kinetic and potential energy equal:

$$K_{i_1} = \frac{1}{8} (m_1 l_1^2 + 4I_{1z}) \dot{\theta}_i^2, \quad U_{i_1} = \frac{1}{2} m_1 g l_1 s\theta_i \quad (28)$$

For $q_5 = \theta$, we have:

$$\begin{aligned}
 & I'_{px} \dot{\psi} c \theta (\dot{\varphi} - \dot{\psi} s \theta) + I'_{py} \dot{\psi} s \varphi s \theta (\dot{\theta} c \varphi + \dot{\psi} s \varphi c \theta) - I'_{py} \dot{\varphi} s \varphi (\dot{\theta} c \varphi + \dot{\psi} s \varphi c \theta) - I'_{pz} \dot{\psi} c \varphi s \theta (\dot{\theta} s \varphi - \dot{\psi} c \varphi c \theta) \\
 & + I'_{py} c \varphi (\dot{\theta} c \varphi - \dot{\varphi} \dot{\theta} s \varphi + \dot{\psi} s \varphi c \theta - \dot{\theta} \dot{\psi} s \varphi s \theta + \dot{\varphi} \dot{\psi} c \varphi c \theta) \\
 & = 2 \sum_{i=1}^6 [s \psi (Y_p - y_{rbi} - l_1 c \phi_{bi} c \theta_i + x_{rpi} c \theta s \psi + y_{rpi} (c \varphi c \psi + s \varphi s \psi s \theta) + z_{rp} (c \varphi s \psi s \theta - c \psi s \varphi)) \\
 & \quad \times (z_{rp} c \varphi c \theta - x_{rpi} s \theta + y_{rpi} c \theta s \varphi) \\
 & \quad - (x_{rpi} c \theta + z_{rp} c \varphi s \theta + y_{rpi} s \varphi s \theta) (Z_p - l_1 s \theta_i - x_{rpi} s \theta + y_{rpi} c \theta s \varphi + z_{rp} c \varphi c \theta) \\
 & \quad + c \psi (X_p - x_{rbi} + l_1 s \phi_{bi} c \theta_i + x_{rpi} c \psi c \theta + y_{rpi} (c \psi s \varphi s \theta - c \varphi s \psi) + z_{rp} (s \varphi s \psi + c \varphi c \psi s \theta)) \\
 & \quad \times (z_{rp} c \varphi c \theta - x_{rpi} s \theta + y_{rpi} c \theta s \varphi)] \lambda_i
 \end{aligned} \tag{34}$$

The i -th rod's kinetic and potential energy are:

For $q_6 = \psi$, we have:

$$K_{i_2} = \frac{1}{4} (I_1^2 m_2 \dot{\theta}_i^2), \quad U_{i_2} = \frac{1}{2} m_2 g l_1 s \theta_i \tag{29}$$

The Lagrange equations are as follows:

For $q_1 = X_p$, we have:

$$m'_p \ddot{X}_p = 2 \sum_{i=1}^6 [X_p - x_{rbi} + l_1 s \phi_{bi} c \theta_i + x_{rpi} c \theta c \psi - y_{rpi} (c \varphi s \psi - s \varphi s \theta c \psi) + z_{rp} (s \varphi s \psi + c \varphi s \theta c \psi)] \lambda_i \tag{30}$$

For $q_2 = Y_p$, we have:

$$m'_p \ddot{Y}_p = 2 \sum_{i=1}^6 [Y_p - y_{rbi} - l_1 c \phi_{bi} c \theta_i + x_{rpi} c \theta s \psi + y_{rpi} (s \varphi s \theta s \psi + c \varphi c \psi) + z_{rp} (c \varphi s \theta s \psi - s \varphi c \psi)] \lambda_i \tag{31}$$

For $q_3 = Z_p$, we have:

$$m'_p (\ddot{Z}_p - g) = 2 \sum_{i=1}^6 [Z_p - l_1 s \theta_i - x_{rpi} s \theta + y_{rpi} s \varphi c \theta + z_{rp} c \varphi c \theta] \lambda_i \tag{32}$$

For $q_4 = \varphi$, we have:

$$\begin{aligned}
 & I'_{px} (\ddot{\varphi} - \ddot{\psi} s \theta - \dot{\theta} \dot{\psi} c \theta) + \left(I'_{pz} - I'_{py} \right) \left[\frac{1}{2} \dot{\psi}^2 s (2 \varphi) (c \theta)^2 - \frac{1}{2} \dot{\theta}^2 s (2 \varphi) + \dot{\theta} \dot{\psi} c (2 \varphi) c \theta \right] \\
 & = 2 \sum_{i=1}^6 [(y_{rpi} c \varphi c \theta - z_{rp} s \varphi c \theta) (Z_p - l_1 s \theta_i - x_{rpi} s \theta + z_{rp} c \varphi c \theta + y_{rpi} s \varphi c \theta) \\
 & \quad + (X_p - x_{rbi} + l_1 s \phi_{bi} c \theta_i + x_{rpi} c \theta c \psi - y_{rpi} (c \varphi s \psi - s \varphi s \theta c \psi) + z_{rp} (s \varphi s \psi + c \varphi s \theta c \psi)) \\
 & \quad \times (y_{rpi} (s \varphi s \psi + c \varphi s \theta c \psi) + z_{rp} (c \varphi s \psi - s \varphi s \theta c \psi)) \\
 & \quad - (Y_p - y_{rbi} - l_1 c \phi_{bi} c \theta_i + x_{rpi} c \theta s \psi + y_{rpi} (c \varphi c \psi + s \varphi s \theta s \psi) - z_{rp} (c \psi s \varphi - c \varphi s \theta s \psi)) \\
 & \quad \times (y_{rpi} (c \psi s \varphi - c \varphi s \theta s \psi) + z_{rp} (c \varphi c \psi + s \varphi s \theta s \psi))] \lambda_i
 \end{aligned} \tag{33}$$

$$\begin{aligned}
& I'_{py}(\dot{\varphi}c\varphi c\theta - \dot{\theta}s\varphi s\theta)(\dot{\theta}c\varphi + \dot{\psi}s\varphi c\theta) + I'_{py}c\theta s\varphi(\ddot{\theta}c\varphi - \dot{\varphi}\dot{\theta}s\varphi + \dot{\psi}s\varphi c\theta - \dot{\theta}\dot{\psi}s\varphi s\theta + \dot{\varphi}\dot{\psi}c\varphi c\theta) \\
& + I'_{pz}(\dot{\varphi}s\varphi c\theta + \dot{\theta}c\varphi s\theta)(\dot{\theta}s\varphi - \dot{\psi}c\varphi c\theta) - I'_{pz}c\varphi c\theta(\ddot{\theta}s\varphi + \dot{\varphi}\dot{\theta}c\varphi - \dot{\psi}c\varphi c\theta + \dot{\varphi}\dot{\psi}s\varphi c\theta + \dot{\theta}\dot{\psi}c\varphi s\theta) \\
& - I'_{px}\dot{\theta}c\theta(\dot{\varphi} - \dot{\psi}s\theta) - I'_{px}s\theta(\ddot{\varphi} - \dot{\psi}s\theta - \dot{\theta}\dot{\psi}c\theta) \\
& = 2 \sum_{i=1}^6 [(Y_p - y_{rbi} - l_1 c\phi_{bi}c\theta_i + x_{rpi}c\theta s\psi + y_{rpi}(c\varphi c\psi + s\varphi s\theta s\psi) - z_{rp}(s\varphi c\psi - c\varphi s\theta s\psi)) \\
& \quad \times (x_{rpi}c\theta c\psi - y_{rpi}(c\varphi s\psi - s\varphi s\theta c\psi) + z_{rp}(s\varphi s\psi + c\varphi s\theta c\psi)) \\
& \quad - (x_{rpi}c\theta s\psi + y_{rpi}(c\varphi c\psi + s\varphi s\theta s\psi) - z_{rp}(s\varphi c\psi - c\varphi s\theta s\psi)) \\
& \quad \times (X_p - x_{rbi} + l_1 s\phi_{bi}c\theta_i + x_{rpi}c\theta c\psi - y_{rpi}(c\varphi s\psi - s\varphi s\theta c\psi) + z_{rp}(s\varphi s\psi + c\varphi s\theta c\psi))] \lambda_i
\end{aligned} \quad (35)$$

For $q_{i+6} = \theta_i$, $i = 1 \dots 6$, we have:

$$\begin{aligned}
Q_{m_i} &= \frac{1}{4}(4I_{1z} + m_1 l_1^2 + 2m_2 l_1^2)\ddot{\theta}_i + C_{vf}\dot{\theta}_i + \frac{1}{2}(m_1 + m_2)gl_1 c\theta_i \\
&+ 2l_1 \{ s\phi_{bi}s\theta_i [X_p - x_{rbi} + l_1 s\phi_{bi}c\theta_i + x_{rpi}c\theta c\psi - y_{rpi}(c\varphi s\psi - s\varphi s\theta c\psi) + z_{rp}(s\varphi s\psi + c\varphi s\theta c\psi)] \\
&- c\phi_{bi}s\theta_i [Y_p - y_{rbi} - l_1 c\phi_{bi}c\theta_i + x_{rpi}c\theta s\psi + y_{rpi}(c\varphi c\psi + s\varphi s\theta s\psi) - z_{rp}(s\varphi c\psi - c\varphi s\theta s\psi)] \\
&+ c\theta_i (Z_p - l_1 s\theta_i - x_{rpi}s\theta + y_{rpi}s\varphi c\theta + z_{rp}c\varphi c\theta) \} \lambda_i
\end{aligned} \quad (36)$$

We obtain the active joint torques Q_i , $i = 1 \dots 6$ by substituting λ_i , $i = 1 \dots 6$ in the six equations of Eq. (36). We combine these equations into one equation. Thus, the Lagrangian IDM is expressed in the standard form as follows:

$$\mathbf{Q}_{Lag} = \mathbf{M}_{Lag} \cdot \ddot{\boldsymbol{\theta}} + \mathbf{V}_{Lag} \cdot \dot{\boldsymbol{\theta}} + \mathbf{g}_{Lag}(\boldsymbol{\theta}) \quad (37)$$

In this form, $\mathbf{M}_{Lag} = (I_{1z} + \frac{m_1}{4}l_1^2 + \frac{m_2}{2}l_1^2) \cdot \mathbf{I}_{3 \times 3}$ is the inertia matrix, $\mathbf{V}_{Lag} = C_{vf} \cdot \mathbf{I}_{3 \times 3}$ is the Coriolis and centrifugal matrix. $\mathbf{g}_{Lag}$ is the gravity vector, such that

$$\begin{aligned}
\mathbf{g}_{Lag} &= \frac{1}{2}(m_1 + m_2)gl_1 c\theta_i \\
&+ 2l_1 \{ s\phi_{bi}s\theta_i [X_p - x_{rbi} + l_1 s\phi_{bi}c\theta_i + x_{rpi}c\theta c\psi - y_{rpi}(c\varphi s\psi - s\varphi s\theta c\psi) + z_{rp}(s\varphi s\psi + c\varphi s\theta c\psi)] \\
&- c\phi_{bi}s\theta_i [Y_p - y_{rbi} - l_1 c\phi_{bi}c\theta_i + x_{rpi}c\theta s\psi + y_{rpi}(c\varphi c\psi + s\varphi s\theta s\psi) - z_{rp}(s\varphi c\psi - c\varphi s\theta s\psi)] \\
&+ c\theta_i (Z_p - l_1 s\theta_i - x_{rpi}s\theta + y_{rpi}s\varphi c\theta + z_{rp}c\varphi c\theta) \} \lambda_i
\end{aligned}$$

4.2 The Newton–Euler inverse dynamic model

In this section, the Newton–Euler method is employed for deriving the IDM of this robot. The same assumptions applied when deriving the Lagrangian model are applied here. The external wrench $\mathbf{F}_{ext,MP}$ (force and torque) acting on the movable plate is transformed to find the torques $\mathbf{Q}_{ext,MP}$ resulted from it by the relation $\mathbf{Q}_{ext,MP} = (\mathbf{J}^T)^{-1} \mathbf{F}_{ext,MP}$ [37]. Hence, the free-body diagram for the i -th crank is as depicted in Fig. 3, where Q_{m_i} is the torque applied by the i -th

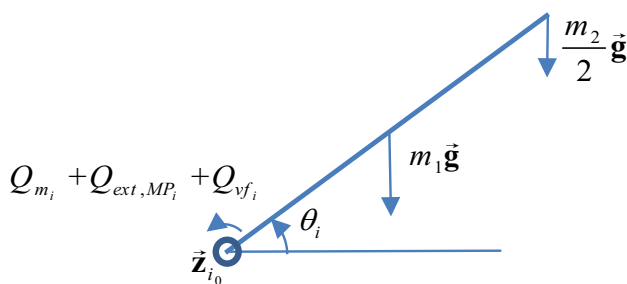


Fig. 3 The external forces and torques exerted on the i -th crank

motor, Q_{vf_i} is the viscous friction torque at the i -th joint, m_1 is the mass of each crank, and m_2 is the mass of each rod.

Accordingly, the Newton's second law for rotation about $\bar{\mathbf{z}}_{i_0}$, which is fixed in the inertial frame B , is written as follows:

$$Q_{m_i} = \left(I_{1z} + \frac{m_1}{4}l_1^2 + \frac{m_2}{2}l_1^2 \right) \ddot{\theta}_i + C_{vf} \cdot \dot{\theta}_i + \frac{m_1 + m_2}{2}gl_1 c\theta_i - Q_{ext,MP_i} \quad (38)$$

By combining the six equations for the six cranks while taking the mapping of the external wrench into consideration, the Newton–Euler IDM is expressed in the standard form as follows:

Table 2 The parameters of the considered robot

Variable	Description	Value
r_b	The base plate's radius	0.272 m
$\phi_{b_i}, i = 1 \dots 6$	The angles of $O_{i_0}, i = 1 \dots 6$ around the base plate	$[60 \ 60 \ 180 \ 180 \ 300 \ 300]^\circ$
$e_i, i = 1 \dots 6$	Displacement of O_{i_0} from the angular position along $\vec{y}_{i_0}$	0.06 m
l_1	The crank's length	0.095 m
l_2	The rod's length	0.253 m
r_p	The movable plate's radius	0.262 m
λ_{p_i}	The angles of $O_{i_1}, i = 1 \dots 6$ around the movable plate	$[60 \ 60 \ 180 \ 180 \ 300 \ 300]^\circ$
$b_{i_1}, i = 1 \dots 6$	Displacement of O_{i_1} from the angular position along $\vec{y}_{i_1}$	0.069 m
z_{rp}	The distance from the movable plate's center of mass to O_{i_1} along $\vec{z}_p$	-0.0232 m
m_1	The crank's mass	0.085 kg
m_2	The rod's mass	0.227 kg
m_p	The movable plate's mass	2.633 kg
I_{1z}	The crank's moment of inertia about an axis parallel to $\vec{z}_{i_0}$ that passes from its center of mass	$68.203 \times 10^{-6} \text{ kg m}^2$
I'_{px}	Moment of inertia of the movable plate with masses of $m_2/2$ condensed at points O_{i_1} about $\vec{x}_p$	$80.702 \times 10^{-3} \text{ kg m}^2$
I'_{py}	Moment of inertia of the movable plate with masses of $m_2/2$ condensed at points O_{i_1} about $\vec{y}_p$	$80.702 \times 10^{-3} \text{ kg m}^2$
I'_{pz}	Moment of inertia of the movable plate with masses of $m_2/2$ condensed at points O_{i_1} about $\vec{z}_p$	$157.784 \times 10^{-3} \text{ kg m}^2$

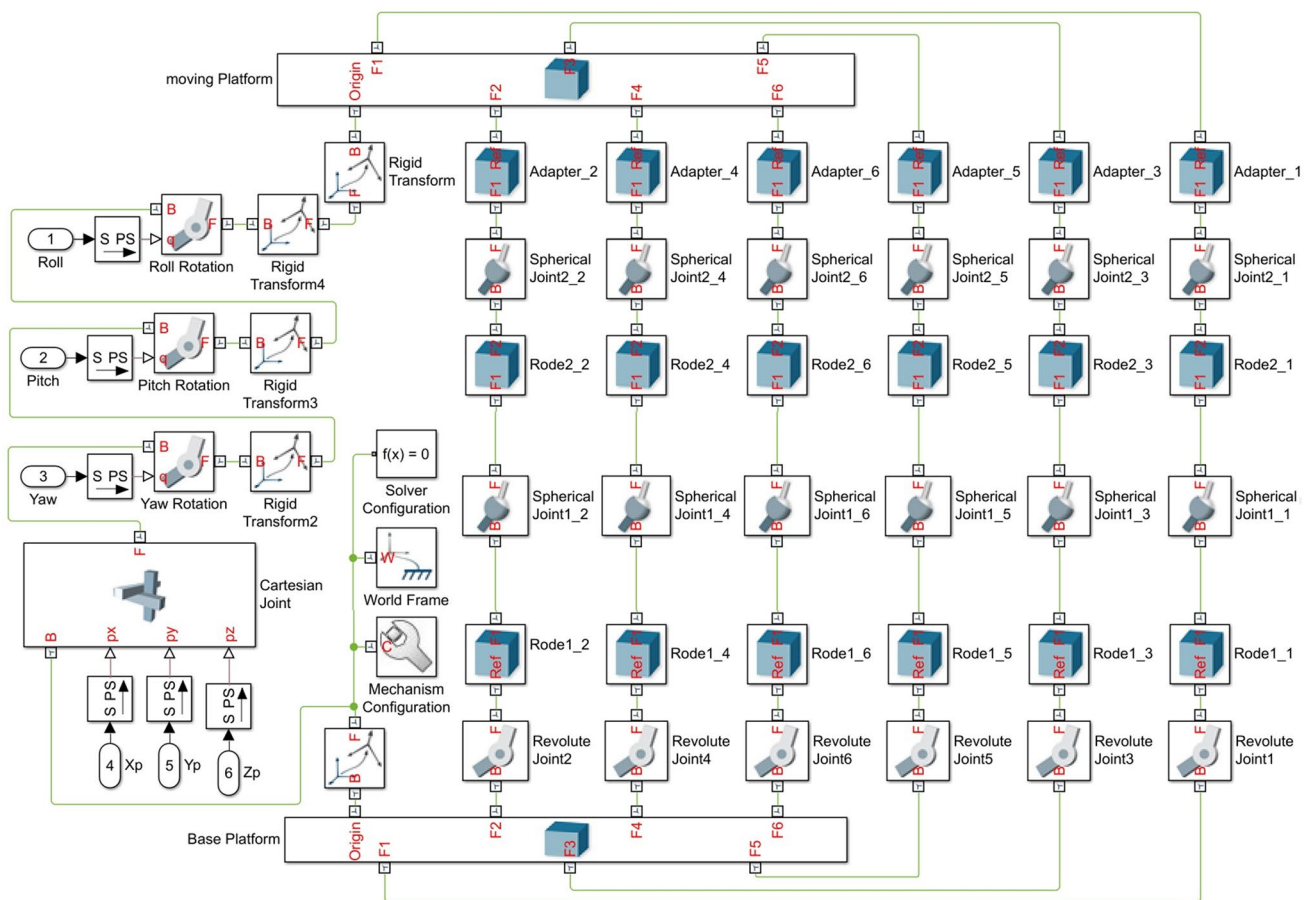
**Fig. 4** The Simscape model of rotary Stewart-Gough robot

Table 3 The relative r.m.s. error in the values obtained from the analytical models for the active joints in comparison with Simscape simulation results

N	Scenario description	$RelRMSE_{\theta}(\%)$	$RelRMSE_{\dot{\theta}}(\%)$	$RelRMSE_{\ddot{\theta}}(\%)$	$RelRMSE_{Q_{Lag}}(\%)$	$RelRMSE_{Q_{NE}}(\%)$
1	$X_p = 0.08 \sin\left(\frac{\pi t}{5}\right)$ $Y_p = 0.08 \left[1 - \cos\left(\frac{\pi t}{5}\right)\right]$ $Z_p = 0.265 + 0.07 \sin\left(\frac{\pi t}{5}\right)$ No rotations	$\begin{bmatrix} 0.01 \\ 0.06 \\ 0.01 \\ 0.05 \\ 0.02 \\ 0.05 \end{bmatrix}$	$\begin{bmatrix} 0.07 \\ 0.07 \\ 0.08 \\ 0.07 \\ 0.07 \\ 0.07 \end{bmatrix}$	$\begin{bmatrix} 0.13 \\ 0.09 \\ 0.16 \\ 0.12 \\ 0.08 \\ 0.13 \end{bmatrix}$	$\begin{bmatrix} 0.28 \\ 0.37 \\ 0.52 \\ 0.47 \\ 0.14 \\ 0.46 \end{bmatrix}$	$\begin{bmatrix} 0.32 \\ 0.39 \\ 0.50 \\ 0.47 \\ 0.20 \\ 0.47 \end{bmatrix}$
2	$\varphi = \frac{\pi}{10} \sin\left(\frac{\pi t}{10}\right)$ $\theta = \frac{\pi}{15} \sin\left(\frac{2\pi t}{15} + \pi\right)$ $\psi = \frac{\pi}{12} \sin\left(\frac{\pi t}{5}\right)$ No translation	$\begin{bmatrix} 0.01 \\ 0.08 \\ 0.01 \\ 0.09 \\ 0.01 \\ 0.07 \end{bmatrix}$	$\begin{bmatrix} 0.09 \\ 0.11 \\ 0.12 \\ 0.11 \\ 0.11 \\ 0.12 \end{bmatrix}$	$\begin{bmatrix} 0.17 \\ 0.21 \\ 0.14 \\ 0.19 \\ 0.19 \\ 0.16 \end{bmatrix}$	$\begin{bmatrix} 0.49 \\ 0.57 \\ 0.52 \\ 0.51 \\ 0.53 \\ 0.50 \end{bmatrix}$	$\begin{bmatrix} 0.62 \\ 0.65 \\ 0.59 \\ 0.62 \\ 0.64 \\ 0.60 \end{bmatrix}$
3	$X_p = 0.06 \sin\left(\frac{\pi t}{10}\right)$ $Y_p = 0.06 \left[1 - \cos\left(\frac{\pi t}{10}\right)\right]$ $Z_p = 0.265 + 0.04 \sin\left(\frac{\pi t}{5}\right)$ $\varphi = \frac{\pi}{30} \sin\left(\frac{\pi t}{6}\right)$ $\theta = \frac{\pi}{20} \sin\left(\frac{2\pi t}{15} + \pi\right)$ $\psi = \frac{\pi}{36} \sin\left(\frac{\pi t}{5}\right)$	$\begin{bmatrix} 0.02 \\ 0.11 \\ 0.02 \\ 0.07 \\ 0.02 \\ 0.11 \end{bmatrix}$	$\begin{bmatrix} 0.13 \\ 0.13 \\ 0.14 \\ 0.13 \\ 0.12 \\ 0.13 \end{bmatrix}$	$\begin{bmatrix} 0.16 \\ 0.15 \\ 0.20 \\ 0.24 \\ 0.13 \\ 0.14 \end{bmatrix}$	$\begin{bmatrix} 0.35 \\ 0.47 \\ 0.53 \\ 0.45 \\ 0.43 \\ 0.49 \end{bmatrix}$	$\begin{bmatrix} 0.44 \\ 0.58 \\ 0.53 \\ 0.40 \\ 0.45 \\ 0.43 \end{bmatrix}$

$$\mathbf{Q}_{NE} = \mathbf{M}_{NE} \cdot \ddot{\boldsymbol{\theta}} + \mathbf{V}_{NE} \cdot \dot{\boldsymbol{\theta}} + \mathbf{g}_{NE}(\boldsymbol{\theta}) \quad (39)$$

In this form, $\mathbf{M}_{NE} = \left(I_{1z} + \frac{m_1}{4}l_1^2 + \frac{m_2}{2}l_1^2\right) \cdot \mathbf{I}_{3 \times 3}$ is the inertia matrix, $\mathbf{V}_{NE} = C_{vf} \cdot \mathbf{I}_{3 \times 3}$ is the Coriolis and centrifugal matrix. $\mathbf{g}_{NE}$ is the gravity vector, where

$$\mathbf{g}_{NE}(\boldsymbol{\theta}) = \frac{m_1 + m_2}{2} g l_1 \begin{bmatrix} c\theta_1 \\ c\theta_2 \\ c\theta_3 \\ c\theta_4 \\ c\theta_5 \\ c\theta_6 \end{bmatrix} - (\mathbf{J}^T)^{-1} \begin{bmatrix} 0 \\ 0 \\ -(m_p + 3m_2)g \\ \sum_{i=1}^6 \overline{BO_p O_{i_7}} \times \begin{bmatrix} 0 \\ 0 \\ -\frac{m_2}{2}g \end{bmatrix} \end{bmatrix} \quad (40)$$

Note that $\mathbf{M}_{Lag} = \mathbf{M}_{NE}$, $\mathbf{V}_{Lag} = \mathbf{V}_{NE}$. These two models differ only in the gravity vector. The numerical validation and assessment of these two models are presented in the next section.

5 Validating the derived models

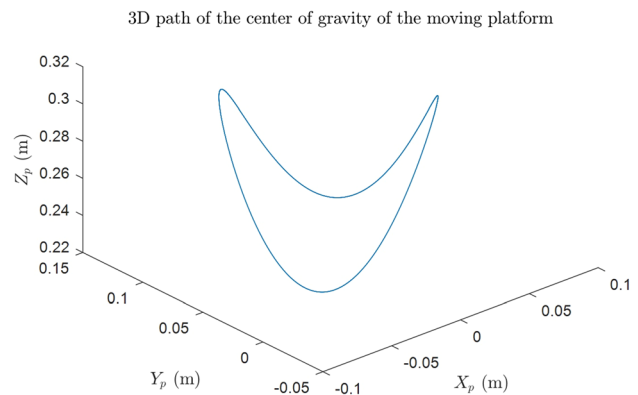
To validate the derived models, we have first developed a 3D model for the robot using Autodesk Inventor® 2018, and then imported the components of the 3D model into Simscape multi-body in the MATLAB® Simulink R2017a

environment and used them to build a Simscape multi-body model for performing the necessary simulations and assessments.

The geometrical and inertial characteristics of the robot are presented in Table 2.

The model built in Simscape is shown in Fig. 4.

The inputs for this model are the linear displacements X_p , Y_p , and Z_p of the center of gravity of the movable plate and Euler angles φ , θ , and ψ representing the rotation of the


Fig. 5 The three-dimensional path of the movable plate's center of mass

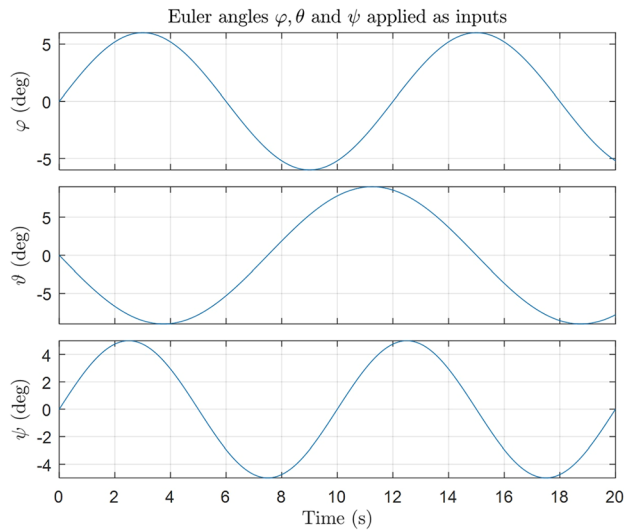


Fig. 6 Euler angles φ , θ and ψ applied as inputs for both Simscape and analytical models

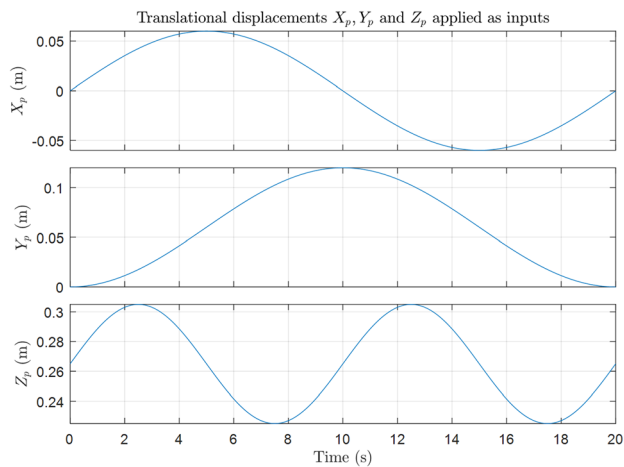


Fig. 7 Translational displacements X_p , Y_p and Z_p applied as inputs for both Simscape and analytical models

movable plate w.r.t. the base plate, along with their first and second derivatives. The outputs of the simulation model, θ_i , $\dot{\theta}_i$, $\ddot{\theta}_i$ and Q_{m_i} for $i = 1 \dots 6$, are compared with the respective values obtained analytically from the IKM, Jacobian and the IDMs for validation purpose. The output blocks are not shown in the figure for maintaining its visibility.

The system has been tested in three scenarios, and the relative root-mean-square error (RMSE) has been used as a criterion for the coincidence between the results of the Simscape multi-body model and the analytical results of the IKM, Jacobian, and the IDMs.

Table 3 shows the relative RMSE values of the errors in the angles, angular rates and angular accelerations of the

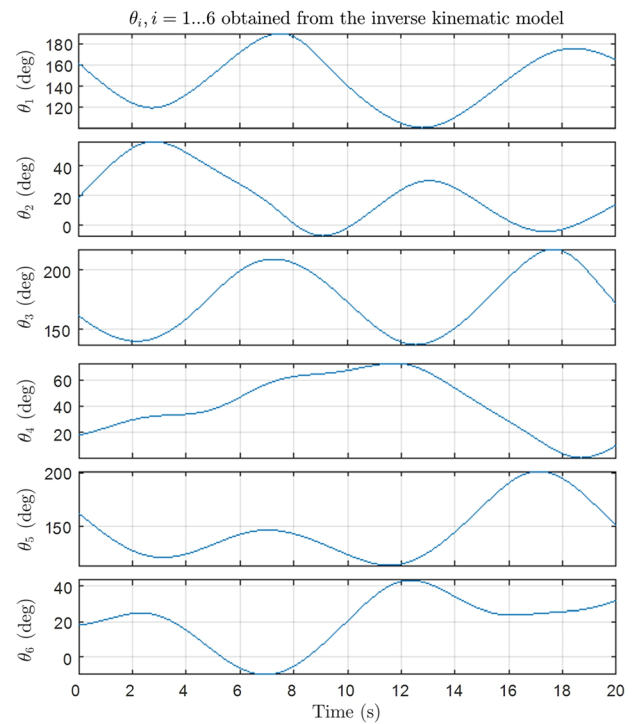


Fig. 8 θ_i , $i = 1 \dots 6$ obtained from the IKM

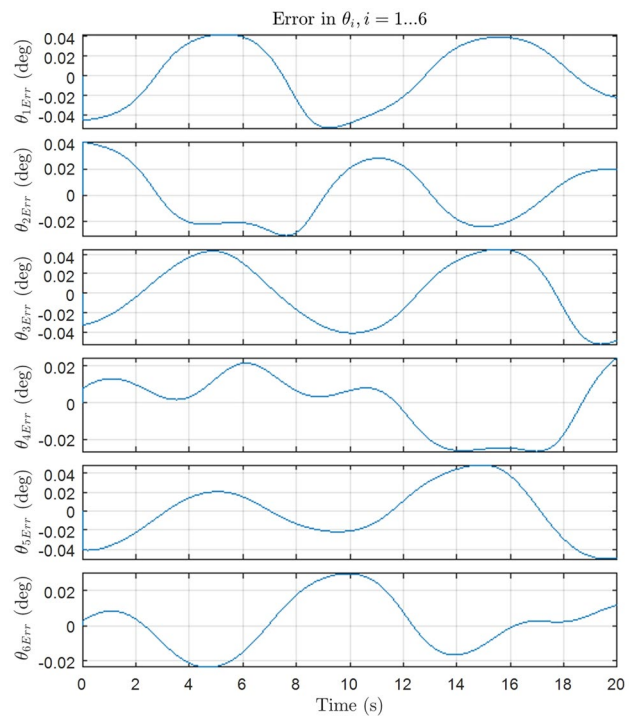


Fig. 9 The error in θ_i , $i = 1 \dots 6$ in comparison with Simscape simulation

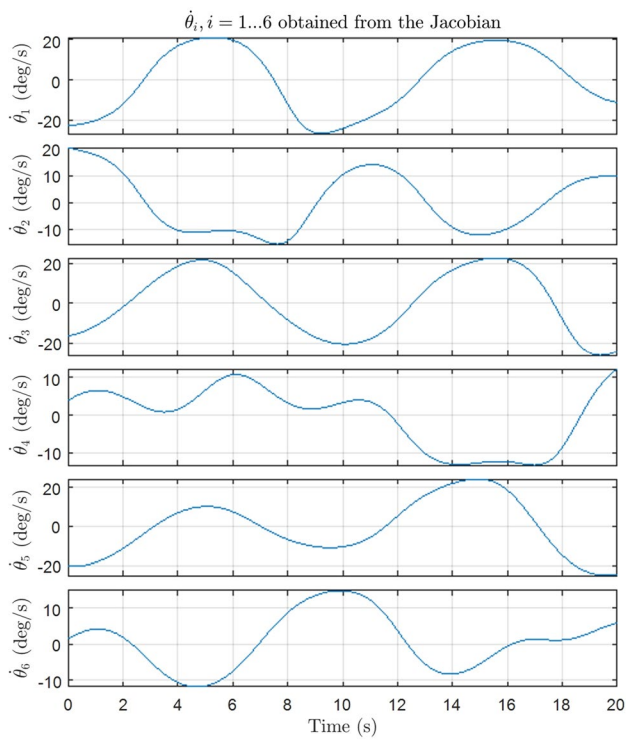


Fig. 10 $\dot{\theta}_i, i = 1 \dots 6$ obtained from the Jacobian

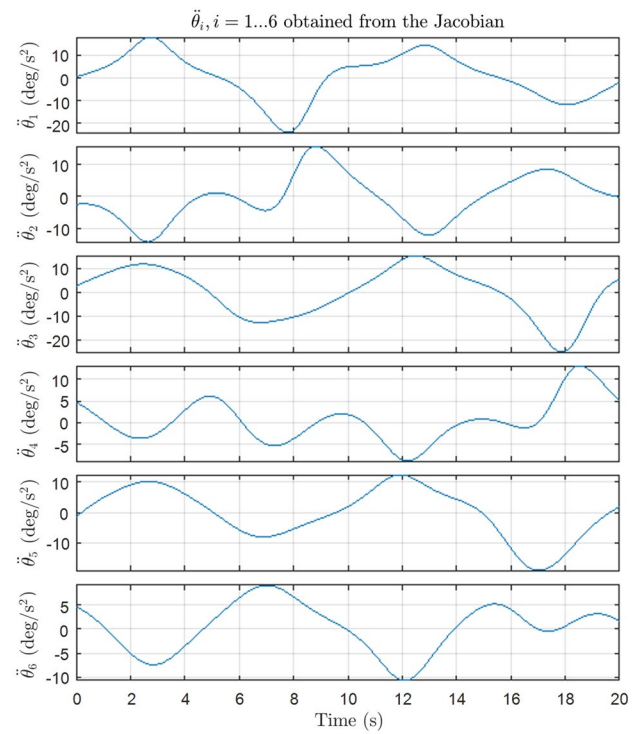


Fig. 12 $\ddot{\theta}_i, i = 1 \dots 6$ obtained from the Jacobian

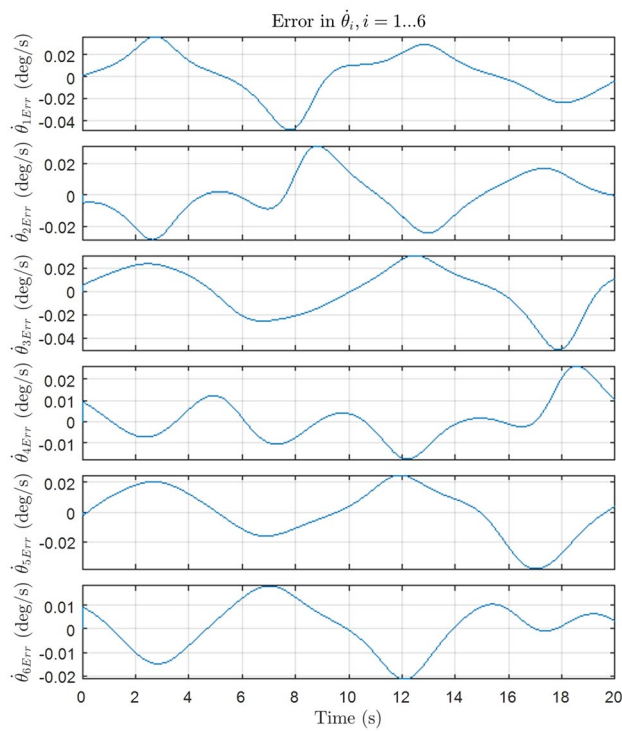


Fig. 11 The error in $\dot{\theta}_i, i = 1 \dots 6$ in comparison with Simscape simulation

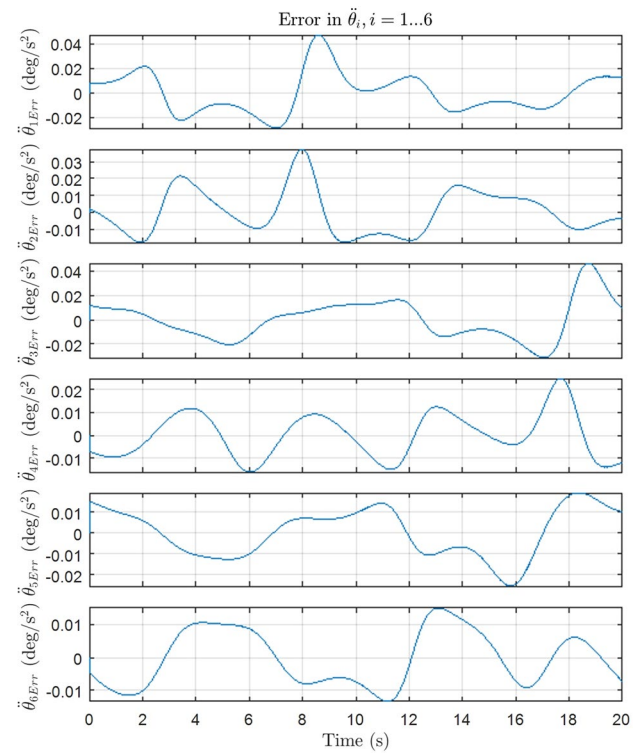


Fig. 13 The error in $\ddot{\theta}_i, i = 1 \dots 6$ in comparison with Simscape simulation

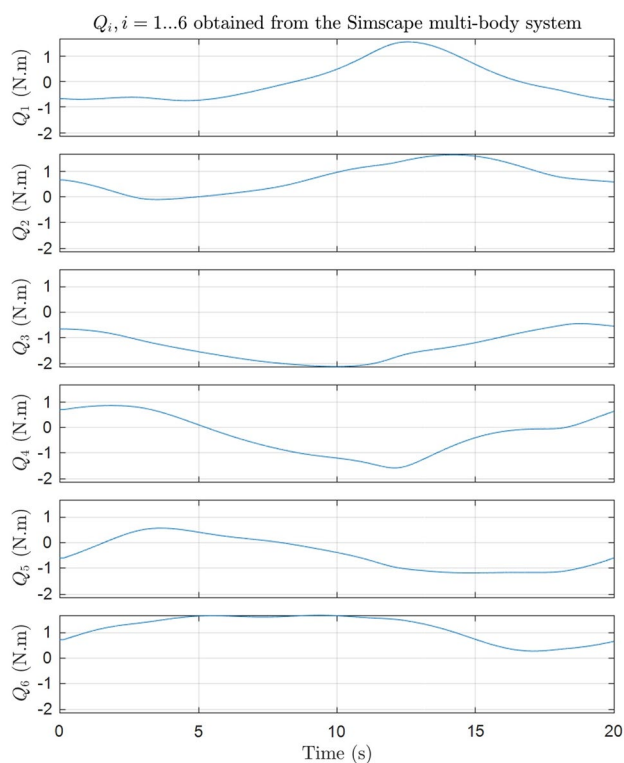


Fig. 14 $Q_i, i = 1 \dots 6$ obtained from Simscape simulation

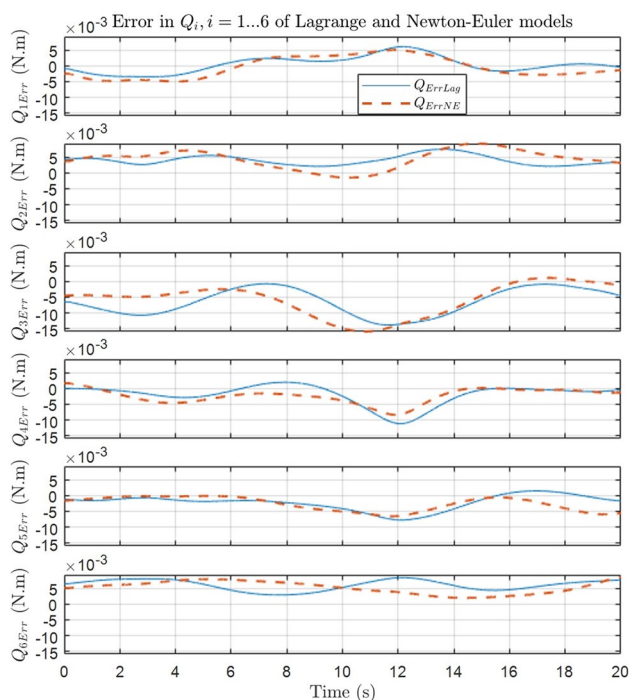


Fig. 15 The error in $Q_i, i = 1 \dots 6$ using Lagrange and Newton–Euler IDMs in comparison with Simscape simulation

active joints and the error in obtaining the torques applied by the active joint motors for three scenarios. The relative RMS values of the errors are calculated as follows:

$$RelRMSE_X(\%) = \frac{\sqrt{\frac{1}{N} \sum_{j=1}^N (X_{M_j} - X_{S_j})^2}}{\sqrt{\frac{1}{N} \sum_{j=1}^N (X_{M_j})^2}} \times 100 \quad (41)$$

in which, X is the considered variable, X_{M_j} is the value of the j -th sample obtained from the analytical model, X_{S_j} is the value of the j -th sample obtained from the simulation, and N is the number of samples.

The first scenario simulates a 3 DOF translational motion of the moving plate. The second scenario simulates a 3 DOF rotational motion of the moving plate. The third scenario simulates a 6 DOF motion of the moving plate. All of these scenarios have been run using the following initial conditions: $X_{P_0} = Y_{P_0} = 0$, $Z_{P_0} = 0.265 \text{ m}$ and $\varphi_0 = \theta_0 = \psi_0 = 0$ with initial values of translational and angular velocities equal to zero.

For visualizing the above results, the time histories of the inputs and outputs of the system in the third scenario are shown in the following figures. The three-dimensional path of the movable plate's center of mass and the Euler angles φ , θ and ψ applied as inputs in the third scenario (which is the more complicated one containing simultaneous translations and rotations of the movable plate) are as shown in Figs. 5 and 6. The translational displacements of the movable plate's center of mass are shown in Fig. 7.

The outputs of the analytical models along with their errors in comparison with the respective outputs obtained from the Simscape simulation are shown in Figs. 8, 9, 10, 11, 12, 13, 14, 15.

As shown in Fig. 9, the error in obtaining $\theta_i, i = 1 \dots 6$ from the IKM is better than 0.05 deg in the 6 DOF motion of the movable plate.

As shown in Fig. 11, the error in obtaining $\dot{\theta}_i, i = 1 \dots 6$ from the Jacobian is better than 0.05 deg / sec in the 6 DOF motion of the movable plate.

As shown in Fig. 13, the error in obtaining $\ddot{\theta}_i, i = 1 \dots 6$ from the Jacobian is better than 0.05 deg / sec² in the 6 DOF motion of the movable plate.

As shown in Fig. 15, the errors in obtaining the torques of the active joints from the Lagrange and Newton–Euler IDMs are better than 0.0138 N.m and 0.0159 N.m, respectively, in the 6 DOF motion of the movable plate. This indicates that the accuracy of both models is acceptable and lies in the same range, meaning that the terms that are different between the two models are numerically equivalent to each

other and that both IDMs can be equivalently used to model the dynamics of this manipulator.

According to Table 3 and Figs. 8, 9, 10, 11, 12, 13, 14, 15, we observe that the derived models were able to provide the angles, angular rates, angular accelerations, and torques of the active joints and the angles and angular rates of the passive joints with a very good accuracy. In all the scenarios, the relative RMS error was better than 0.12% for the active joint angles, better than 0.15% for the active joint angular rates, better than 0.25% for the active joint angular accelerations, and better than 0.57% and 0.65% for the active joint torques using the Lagrange and Newton–Euler IDMs, respectively.

The reason for the higher relative RMS error in obtaining the torque values is due to the approximation applied for the mass distribution of the rods when deriving both IDMs.

Another noteworthy point about these two IDMs is that the Newton–Euler method results in a closed form model while in the Lagrangian model, the coefficients λ_i must be first calculated, and then their numerical values are used in the final form. This closed form makes the Newton–Euler model more suitable to be used in linear control systems because it allows this model to be linearized around a desired work point.

To compare the execution time, both IDMs were run for more than 500,000 times in MATLAB R2017a under Windows 7 on a laptop with Intel Core i7-Q480@1.87 GHz processor and 10 GB of RAM. The mean execution time was 382 μ s for Lagrange IDM and 237 μ s for Newton–Euler IDM, indicating that the calculations of the Newton–Euler model are faster than the calculations of the Lagrangian model in the simulation environment. This indicates that the Newton–Euler model will probably be faster in real time applications and more suitable to be implemented in them.

6 Conclusions

In this paper, the 6-RSS Stewart–Gough robot configuration was investigated. The inverse kinematic model (IKM) and two inverse dynamic models (IDMs), namely the Lagrange and the Newton–Euler models, were derived and validated for this manipulator. The models were validated by comparing their results with the results of the model built using Simscape multi-body in the MATLAB Simulink environment for three scenarios: 3 DOFs translation, 3 DOF rotation and 6 DOF translational and rotational motion of the moving plate. The results show that the relative RMS error was better than 0.12% for the active joint angles, better than 0.15% for the active joint angular rates and better than 0.25% for the active joint angular accelerations. This shows a very good accuracy of the IKM as compared to the Simscape model. The relative RMS for the active joint torques using the

Lagrange and Newton–Euler IDMs was better than 0.57% and 0.65%, respectively. This also shows that both inverse dynamic models have a very good coincidence with each other and with the simulation outputs. Although both IDMs provide similar results, the Newton–Euler approach has two advantages over the Lagrangian approach. The first is that it results in an explicit analytical closed form model while the Lagrange modeling method results in a model that should be solved numerically. This makes the Newton–Euler model more appropriate to be analytically processed such that it can be linearized and employed in control system strategies where is necessary the model linearization. The second advantage of Newton–Euler model is that its calculations are faster than the other model in the simulation environment making it probably more eligible to be implemented in real time applications. The present work forms an initial step for future research on the modeling, model-based control, and parameter optimization of this robot configuration.

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